



proximity of the airport and diverting the threat away from the targeted aircraft. Fig. 10 illustrates the operational concept of Vigilant Eagle airport protection system.

RANETS-E. Ranets-E, described as radio frequency cannon, is an HPM weapon system designed to protect a terminal by disrupting guidance systems of precision-guided munitions and avionics systems of aircraft. The system is configured around an X-band HPM source generating peak power output of 500 megawatts. Pulsed output of 10- to 20-nanosecond pulses at a repetition rate of 500Hz produces an average power of 2.5 kilowatts to five kilowatts. The rooftop steerable parabolic antenna has a gain of 45dB to 50dB.

RANETS-E weapon system reportedly has a lethal range of about 30 kilometres against the electronic guidance systems of precision-guided munitions and aircraft avionics systems, assuming vulnerability threshold field strength of about 1kV/m. This makes RANETS-E a credible lethal weapon against electronic systems at ranges typical of a terminal-point defence weapon.

HPM BLACKOUT. BAE Systems Bofors HPM Blackout is a mobile gigawatt-level microwave source operating in L-band. The system comprises an HPM source, an integrated pulsed-power unit and an exchangeable conical horn antenna with support systems including a compact battery-powered vacuum system for the microwave tube and a gas supply system for the pulsed-power unit. The integrated battery unit renders the system operational in all terrains.

Bofors HPM Blackout, originally designed as a research and evaluation tool for generating data on threats from electromagnetic effects of HPM weapons, could have opera-

tional capabilities with its proven destructive effects at considerable distances against a wide range of commercially-off-the-shelf equipment.

SEPS. This system, designed by Whittaker Corp. in Simi Valley, California, USA, is a portable radio frequency (RF) jammer that can be programmed to jam a specific range of frequencies. The system offers effective electronic countermeasures against electronic proximity fuses used in incoming artillery and mortar rounds by countering the threat of RF-fused munitions by initiating pre-

While this system defeats proximity-fused munitions, weapons with impact fuses go unchallenged. Based on the extensive live fire testing it has been subjected to, the system has demonstrated the ability to significantly enhance survivability of troops and high-value assets from indirect fire proximity-fused munitions.

Warlock Green and Warlock Red. Warlock Green and Warlock Red designed by EDO Communications and Countermeasures (now ITT Electronic Systems, EWS), too, are electronic countermeasure systems that are modified versions of battle-proven SEPS designed by Whittaker Corp. Warlock systems have been found to be effective in countering the threat of radio-controlled IEDs.

An RF signal emitted by Warlock systems jams the communications signals used to detonate the IEDs. It does so by intercepting the signal sent

from a remote station to initiate detonation and disallowing to make contact with the IED. To be effective, one needs to transmit the right frequency, which may not be easy.

Another application of Warlock systems is to disrupt communication devices used by the enemy. Warlock Red is a less sophisticated low-cost version of Warlock Green and is designed to counter certain specific threats that are used in large numbers. Multiple Warlock systems may be used to provide area coverage without causing any mutual interference. Warlock systems have undergone many revisions to add more frequencies and better software.

DS-110 High-Power Electromagnetic (HPHEM) System. DS-110 developed by Diehl Munitions Systems of Germany is a high-power RF system that generates wideband EMP

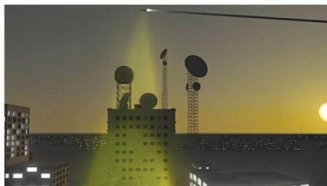


Fig. 11: Illustration of HPM attack by CHAMP

mature detonation. Proximity fuses of artillery and mortar shells use radar signals to compute the distance from target and are programmed to generate fuse commands at the pre-programmed distance.

The system receives the radar signal and then retransmits a modified signal to make the shell believe it has reached programmed distance while it actually has not. Premature detonation offers protection to friendly ground troops, vehicles, structures and other equipment under fire.

The system is also capable of effectively defeating remotely-controlled improvised explosive devices (IEDs) or suicide bombers. Packaged in a case similar in size to a suitcase, Shortstop's passive electronics and operational features make it immune to detection by enemy sensors.